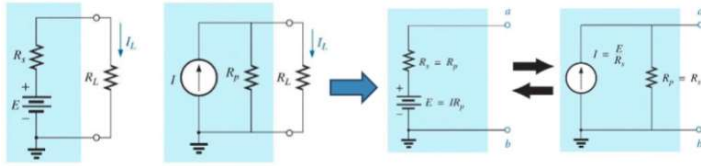


Current Source and Voltage Source Conversion



To Find Current

Branch Current Method:

- Assign a current at each branch of a network.
- Indicate polarities of the resistor (apply Fact 2).
- Write one KVL equation for every closed, independent loop, then solve them in matrix form.

Mesh Analysis:

Fact 1: Mesh Analysis will have 2 polarities when shared by 2 loops. Useful for networks with **voltage sources**.

Fact 2: For polarity, the current's "upstream" end of a resistor is always +, and the "downstream" end is -

Fact 3: If there is a **current source**, remove the current source. The resulting path becomes a **supermesh** current. Setup 1 KVL equation for the supermesh and another equation relating the relevant currents equal to the current source.

MA general approach steps:

1. Assign CW current loop (I_1, I_2 , etc.) at each "window," or closed loop of a network.
2. Indicate polarities determined by the direction of the loop current.
3. Write one KVL equation for every closed, independent loop then solve them in matrix form.

Mesh Analysis (**Shortcut**, Format Approach):

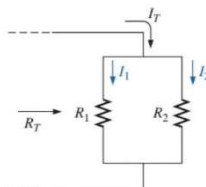
1. Assign currents, same as general approach.
2. Setup blank $n \times n$ matrix where $n = \#$ of nodes for current.
3. Find resistance matrix elements, polarity is not considered.
 - Diagonal elements: $R_{ii} = \sum R_{\text{mesh}}$
 - Other elements: $R_{ij} = -R$ shared by loop i and j (make sure it's negative)
 - Note: $R_{ij} = R_{ji}$
4. $[\text{Matrix}][\text{currents}] = [\text{loop total voltage}]$, then rref

Voltage Divider (To find V through a resistor)

$$V_{out} = V_{in} \left[\frac{R_2}{R_1 + R_2} \right] \quad V_1 = \frac{R_1 E}{R_T} = \frac{R_1 E}{R_1 + R_2 + R_3}$$

Current Divider (To find I through a resistor)

$$I_x = \frac{R_T}{R_x} I_T \quad I_x = \frac{R_1}{R_1 + R_2} I_T$$



To Find Voltage, Current too ig

Nodal Analysis:

Fact 1: Useful for networks with **current sources**

NA general approach steps:

Fact 2: If there is a **voltage source**, replace the voltage source and any component connected to it with short circuit equivalents and apply KCL to the defined nodes.

1. Place reference node at known V (usually at ground where $V=0V$)
2. Assign voltage to essential nodes (V_1, V_2 , etc.)
3. Assign current to each branch
4. Write KCL equations for each node (current in = current out) – ex: $3 = i_1 + i_2$
5. Write each current as an equation using Ohm's Law ($i_1 = \text{voltage drop or gain}/R$) – ex: $(V_1 - 0)/2, V_1 - V_2/10$
6. Substitute current equation with KCL equation then solve for unknown voltages
7. (optional) Plug voltages to step 4 equations to get current

Nodal Analysis with **Supernode**:

1. Do steps 1, 2, 3 same as general approach
2. Replace entire voltage source (and resistor if it's only connected to the source) into a super node
3. Treat supernode as one node and write one KCL equation for it (all currents in = all currents out)
4. Do step 5,6,7 same as general approach

Nodal Analysis (**Shortcut**, Format Approach):

1. Same as step 1 general approach
2. Same as step 2 general approach
3. Setup blank $n \times n$ matrix where $n = \#$ of nodes for voltage.
4. Apply KCL with only the current sources directly connected to the node for the RHS of matrix (aka 6A in, 4A out, $V_1 = 2A$)
5. Find conductance matrix elements using resistors
 - Diagonal Elements: $G_{ii} = \text{sum of } G \text{ connected to each node}$
 - Other elements: $G_{ij} = -G$ of resistor between 2 node i and node j (make sure it's negative!)
 - Note: $G_{ij} = G_{ji}$
6. $[\text{Matrix}][\text{voltages}] = [\text{node current}]$, then rref

Definition:

Node - Place where two or more components connect

Branch - Path between two nodes

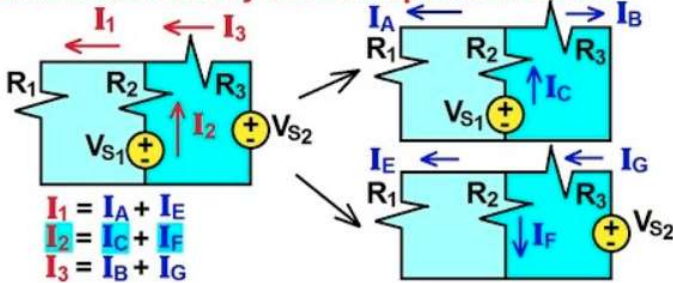
Essential node - Place where three or more components connect

Essential branch - Path which connects two essential nodes

Superposition Defined:

Superposition Defined

The voltage across or the current through an element in a linear circuit is the algebraic sum of that contributed by each independent source.



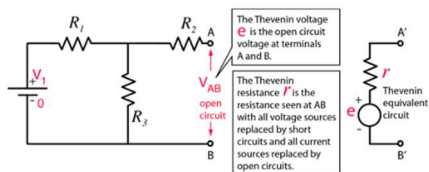
To Solve a Circuit w/ Superposition:

1. Rewrite the circuit by splitting it with independent sources like that pic above
2. Add voltage essential nodes on both new circuits
3. Simplify both circuits using equivalent resistors
4. Find $I_T = V/R_T$ on Circuit 1
5. Find V of Node 1 by doing $V_1 = E - (I_T * R_x)$
6. Find I of resistor using current divider now that you have I_T from step 4
7. Repeat step 4,5,6 for all the independent circuits
8. Sum up all V_s and I_s found from independent circuits to find the main circuit values

Note: just like kinda split em up and sum everything lol, it's super situational

Thevenin Theorem:

Definition: Any combination of voltage source, current source, and resistors with two terminals can be replaced by a single voltage source e and a single series resistor r (aka R_{TH}). The value of e (aka V_{TH}) is the open circuit voltage at the terminals, and the value of r is e divided by the current with the terminals short circuited.



To Find R_{TH} : Remove current sources and short voltage sources

To Find V_{TH} : Remove R_L and find V across open

Series Circuits

Definition: R_1 and R_2 connected, resulting in one and only one connection with each other.

☆ **Fact 4:** All currents are the same through all elements in series.

Fact 5: The total resistance is unaffected by the order in which resistors in series are connected.

Fact 6: In series, batteries can be rearranged.

Current in Series: $I_s = I_1 = I_2 = I_3$

Total Current in Series: $I_s = \frac{V}{R_1 + R_2 + R_3} = \frac{V}{R_T}$

Total Resistance in Series: $R_T = R_1 + R_2 + R_3$

Total Resistance in Series: $R_T = \frac{E}{I}$

Voltage Sources in Series: $E_T = E_1 + E_2 + E_3$

Parallel Circuits

Definition: Two elements, branches, or circuits are in parallel if they have two points in common.

☆ **Fact 7:** All voltages are the same through all elements in parallel.

Voltage in Parallel: $V_s = V_1 = V_2 = V_3$

Total Resistance in Parallel: $R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}}$

Special Case: $R_T = \frac{R_1 R_2}{R_1 + R_2}$ when $\frac{1}{R_1} + \frac{1}{R_2}$

Conductance: $G = G_1 + G_2 + \dots + G_N$, where $G = 1/R$

Total Current in Parallel: $I_s = \frac{V}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}}$

Branch-current method A technique for determining the branch currents of a multiloop network.

Bridge network A network configuration typically having a diamond appearance in which no two elements are in series or parallel.

Current sources Sources that supply a fixed current to a network and have a terminal voltage dependent on the network to which they are applied.

Delta (Δ), pi (π) configuration A network structure that consists of three branches and has the appearance of the Greek letter delta (Δ) or pi (π).

Determinants method A mathematical technique for finding the unknown variables of two or more simultaneous linear equations.

Mesh analysis A technique for determining the mesh (loop) currents of a network that results in a reduced set of equations compared to the branch-current method.

Mesh (loop) current A labeled current assigned to each distinct closed loop of a network that can, individually or in combination with other mesh currents, define all of the branch currents of a network.

Nodal analysis A technique for determining the nodal voltages of a network.

Node A junction of two or more branches in a network.

Supernode A current defined in a network with ideal current sources that permits the use of mesh analysis.

Supernode A node defined in a network with ideal voltage sources that permits the use of nodal analysis.

Wye (Y), tee (T) configuration A network structure that consists of three branches and has the appearance of the capital letter Y or T.

Maximum power transfer theorem A theorem used to determine the load resistance necessary to ensure maximum power transfer to the load.

Millman's theorem A method using source conversions that will permit the determination of unknown variables in a multiloop network.

Norton's theorem A theorem that permits the reduction of any two-terminal linear dc network to one having a single current source and parallel resistor.

Reciprocity theorem A theorem that states that for single-source networks, the current in any branch of a network, due to a single voltage source in the network, will equal the current through the branch in which the source was originally located if the source is placed in the branch in which the current was originally measured.

Substitution theorem A theorem that states that if the voltage across and current through any branch of a dc bilateral network are known, the branch can be replaced by any combination of elements that will maintain the same voltage across and current through the chosen branch.

Superposition theorem A network theorem that permits considering the effects of each source independently. The resulting current and/or voltage is the algebraic sum of the currents and/or voltages developed by each source independently.

Thevenin's theorem A theorem that permits the reduction of any two-terminal, linear dc network to one having a single voltage source and series resistor.